ESARAPU VARUN

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SUMMARY

Electronics and Communication Engineering graduate with experience in Python, machine learning, and backend development using Django REST Framework. Built end-to-end applications involving API design, data processing, and system workflows.

Skilled in TensorFlow, scikit-learn, Pandas, and SQL with strong problem-solving and communication abilities.

WORK EXPERIENCE

A I & M L Developer Apprentice

D SQUARE CONSULTING SERVICES PVT LTD | May 2026 - Nov 2026

- Collaborated with the AI/ML team to develop, test, and optimize supervised and unsupervised machine learning models for predictive modeling and clustering tasks.
- Engineered end-to-end AI pipelines, taking ownership of data ingestion, prototyping with emerging AI tools, and assisting in deploying models onto cloud platforms.
- Participated actively in code reviews, debugging sessions, and model optimization to maintain high code quality standards.
- Documented project workflows and continuously evaluated emerging AI frameworks to drive innovation in a fast-paced development environment.

A I Operator Intern

XTRANSMATRIX CONSULTING SERVICES PVT LTD | Nov 2025 - Mar 2026

- Managed Human-in-the-Loop (HITL) data annotation and intent classification for a US-based healthcare AI project using real-world datasets.
- Validated training pipelines by labeling data and ensuring high-quality inputs for production-level machine learning models.
- Reviewed and evaluated LLM-generated outputs to systematically identify edge cases, inconsistencies, and hallucinations for model refinement.
- Ensured reliability and accuracy across production AI systems by strictly adhering to standardized operating procedures (SOPs).

SKILLS SUMMARY

Languages : Python,C, SQL.
Databases : MySQL, MongoDB.
Backend : Django , Django REST Framework
Machine Learning / Data Science : Pandas, NumPy, scikit-learn, Matplotlib, TensorFlow, Keras, PyTorch.
Tools : Git**,** Jupyter Notebook, Visual Studio Code, IntelliJ IDEA.
Cloud : AWS (basic : EC2 , S3)
Soft Skills : Rapport Building, People Management, Public Speaking.
Concepts : Data Preprocessing, Deep Learning, NLP, CNN, LSTM, Recommendation Systems

EDUCATION

HKBK College Of Engineering | Bangalore, KA | Dec 2022 - Jul 2026

B.E. in Electronics and Communication Engineering

GPA : 8.71/10

PROJECTS

Cinematch : Distributed Movie Recommendation System | Website Link | GitHub Link | October 2026
Engineered an end-to-end distributed collaborative filtering pipeline on **32M+ MovieLens interaction records** using **PySpark 3.5 MLlib**

ALS . Architected a decoupled microservice serving boundary using **Django REST Framework** and **PostgreSQL 15** with indexed composite keys, delivering **sub-15ms inference latency**. Developed a reactive **Streamlit** frontend integrated with the **TMDB Live Graph API** for real-time 4K backdrop enrichment, director chronologies, and continuous tab navigation.

Vehicle Service Management System | LINK | Apr 2026

Built **Backend RESTful APIs** using **Django REST Framework** to manage vehicles, components, issues, and payments. Implemented **issue lifecycle management** with **status transitions** and **validation logic**.
Designed **cost calculation** combining labor and component pricing. Developed a **Streamlit interface** for updating status, simulating payments, and visualizing revenue.

Real-time Speech Emotion Recognition system | LINK | Oct 2025

Extracted **MFCC** features using **Librosa**. Applied **preprocessing** including **normalization** and **padding**. Designed and trained a **CNN-based machine learning model** for classification.
Performed **model evaluation** and **optimization** to improve performance and accuracy

Text Emotion Classification Model using LSTM | LINK | Aug 2025

Built an **LSTM-based NLP model for emotion classification with 87% accuracy** using TensorFlow/Keras. Handled text preprocessing (tokenization, padding, normalization) on custom datasets.
Utilized Python libraries including **Pandas**, **NumPy**, **scikit-learn**, **TensorFlow**, and **Keras** for model building and evaluation

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CERTIFICATES

Python Programming (SoloLearn) **| CERTIFICATE | Apr 2022**

Scientific Computing with Python (FreeCodeCamp) **| CERTIFICATE | Jul 2022**

Machine Learning with Python (FreeCodeCamp) **| CERTIFICATE | Aug 2022**

Architecting SAP on AWS (Amazon Web Services) **| CERTIFICATE | Nov 2025**

EXTRA-CURRICULAR ACTIVITIES

- Hosted and coordinated multiple **college fests**, managing logistics, sponsors, and student teams for smooth execution
- Organized domain-specific **ECE workshops (VACs)** with 60+ participants, ensuring effective content delivery and engagement
- Served as **Event Coordinator** for Technisium 2025, overseeing 8 technical events with **150+ participating teams** from all over Karnataka.